

BRETON PLOTS, AB, Canada

WMO: 712330

Lat: 53.0894N

Lon: 114.4408W

Elev: 852

StdP: 91.51

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.9	-26.3	-34.9	0.2	-29.4	-30.8	0.2	-25.6	7.3	-2.0	6.4	-2.4	1.1	150	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	28.0	17.5	26.0	16.4	24.3	15.5	19.5	25.5	18.0	23.6	16.7	22.4	3.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	13.7	22.3	15.9	12.5	21.0	14.6	11.5	19.2	59.4	25.7	54.4	23.8	50.3	22.6	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.0	6.0	5.1	-33.8	31.2	3.4	2.5	-36.2	33.0	-38.2	34.5	-40.1	35.9	-42.6	37.7
			-33.6	21.3	3.1	1.9	-35.8	22.7	-37.7	23.8	-39.4	24.9	-41.7	26.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-8.6	-6.7	-2.9	4.1	9.9	13.8	16.5	15.3	10.7	4.7	-2.7	-7.9
	DBStd	10.82	9.33	7.82	7.54	5.28	4.43	3.15	3.15	3.23	4.31	4.76	6.72	8.28
	HDD10.0	2848	576	469	400	185	57	4	1	2	42	174	382	556
	HDD18.3	5303	834	702	658	428	263	140	74	104	230	423	632	814
	CDD10.0	620	0	0	1	7	53	119	202	166	63	9	1	0
	CDD18.3	33	0	0	0	0	1	4	18	9	1	0	0	0
	CDH23.3	595	0	0	0	2	33	81	247	173	53	6	0	0
	CDH26.7	131	0	0	0	0	3	17	62	39	9	1	0	0
Wind	WSAvg	2.1	1.8	1.9	2.3	2.7	2.6	2.2	2.0	1.9	1.9	2.1	1.9	1.8
Precipitation	PrecAvg	491	25	16	19	31	55	87	94	63	39	24	21	17
	PrecMax	652	75	30	36	55	110	175	208	138	87	76	61	47
	PrecMin	321	2	1	3	5	18	25	46	31	9	5	4	5
	PrecStd	79	17	8	8	15	23	34	39	21	22	17	15	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	11.4	14.3	22.2	26.4	29.1	31.3	30.2	27.8	22.7	15.0	10.4
		MCWB	4.3	3.9	6.0	10.6	13.2	17.6	20.1	18.2	15.7	12.2	7.4	4.3
	2%	DB	8.0	8.1	11.0	19.0	23.9	25.5	28.3	27.3	24.6	17.7	10.9	7.5
		MCWB	2.8	2.4	4.6	9.1	12.5	15.6	18.7	17.2	14.6	9.9	5.0	2.8
	5%	DB	5.7	5.6	8.7	16.0	21.6	23.2	26.0	25.1	22.0	14.7	8.0	5.0
		MCWB	1.4	0.8	3.1	7.3	11.5	14.7	17.4	16.5	13.4	8.0	3.4	1.0
	10%	DB	3.3	3.3	6.5	13.2	19.1	21.3	24.0	23.0	19.2	12.3	5.4	2.7
		MCWB	0.0	-0.5	1.9	5.9	10.3	13.6	16.4	15.5	12.2	6.8	1.8	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	4.5	6.5	11.5	14.9	19.0	22.0	20.6	16.9	13.3	8.1	4.8
		MCDB	9.7	10.5	12.7	20.1	23.7	27.1	29.3	26.6	24.1	21.4	14.1	9.7
	2%	WB	3.2	2.7	4.7	9.6	13.2	17.1	20.1	18.9	15.4	10.6	5.4	2.8
		MCDB	7.5	7.7	10.2	18.2	21.5	23.3	26.4	24.9	22.7	16.6	10.0	6.9
	5%	WB	1.7	1.2	3.5	7.8	11.9	15.8	18.6	17.6	14.1	8.8	3.8	1.2
		MCDB	5.3	5.1	8.1	15.0	19.7	21.4	24.0	23.4	20.5	13.8	7.6	4.6
	10%	WB	0.2	-0.2	2.1	6.2	10.8	14.6	17.2	16.3	12.7	7.3	2.0	-0.5
		MCDB	3.2	2.9	6.1	12.4	18.0	19.9	22.6	21.6	18.2	11.5	4.8	2.2
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.5	10.9	11.5	12.5	11.8	12.3	12.5	12.5	10.9	9.3	9.6
		MCDBR	11.9	13.7	13.6	16.7	16.8	15.6	15.6	16.3	18.2	15.2	12.1	10.6
	5% WB	MCWBR	7.8	9.0	8.1	8.8	7.7	7.3	7.4	7.9	9.0	8.5	7.5	7.2
		MCWBR	11.6	13.2	13.1	16.0	15.3	13.5	14.4	14.9	16.4	14.6	11.5	10.4
Clear-Sky Solar Irradiance	taub		0.250	0.257	0.296	0.328	0.342	0.350	0.350	0.357	0.311	0.281	0.253	0.239
		taud	2.341	2.390	2.393	2.381	2.387	2.384	2.431	2.368	2.492	2.560	2.475	2.325
	Ebn at Noon		759	874	900	907	907	900	893	865	874	840	762	701
		Edh at Noon	56	73	91	106	111	113	106	107	83	62	49	47
All-Sky Solar Radiation	RadAvg		0.98	1.96	3.40	4.62	5.51	5.86	6.08	5.04	3.51	2.05	1.05	0.72
	RadStd		0.07	0.14	0.21	0.40	0.39	0.41	0.37	0.34	0.37	0.22	0.09	0.06

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		-0.82	N/A	N/A	-1.56	-2.20	-2.16	N/A	+281	-70	-27
Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page